ĐẠI HỌC QUỐC GIA TP. HỒ CHÍ MINH
TRƯỜNG ĐẠI HỌC BÁCH KHOA

GIẢI TÍCH 2

NHẬN DẠNG MẶT BẬC HAI

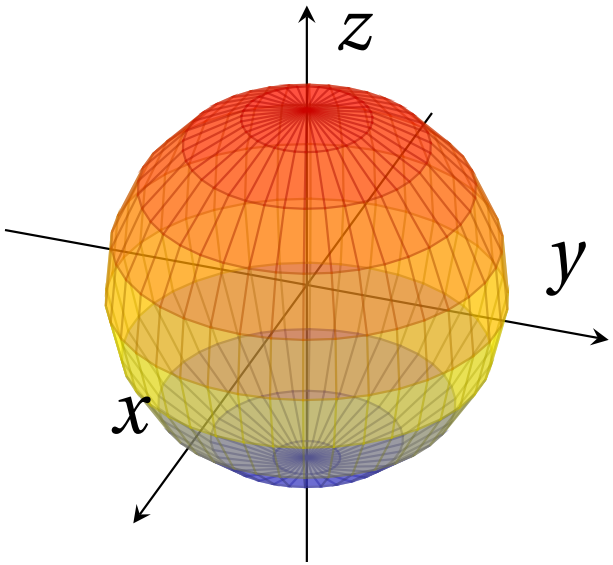
Phương trình tổng quát của mặt bậc hai

$$Ax^2 + By^2 + Cz^2 + Dxy + Exz + Fyz + Gx + Hy + Iz + K = 0$$

Điều kiện:

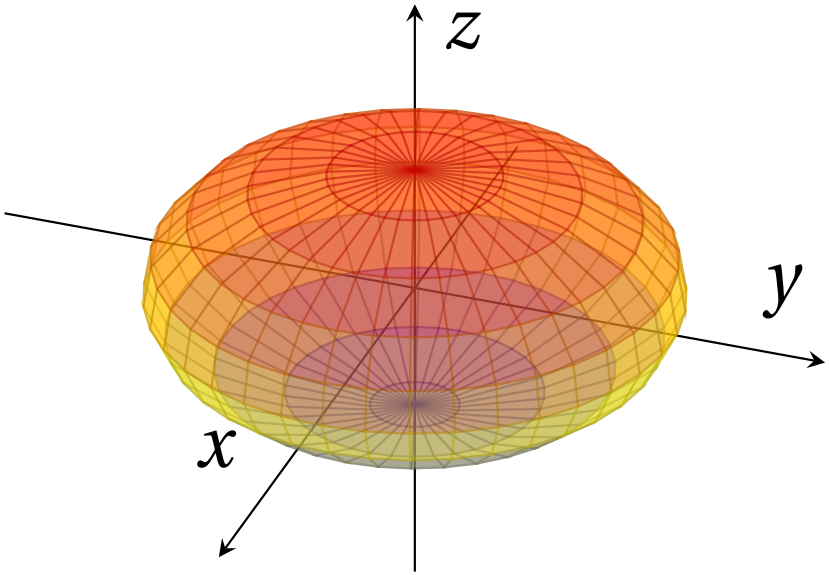
$$A^2 + B^2 + C^2 \neq 0$$

Mặt cầu



$$x^2 + y^2 + z^2 = R^2$$

Mặt Ellipsoid



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$$

Mặt trụ Ellipse

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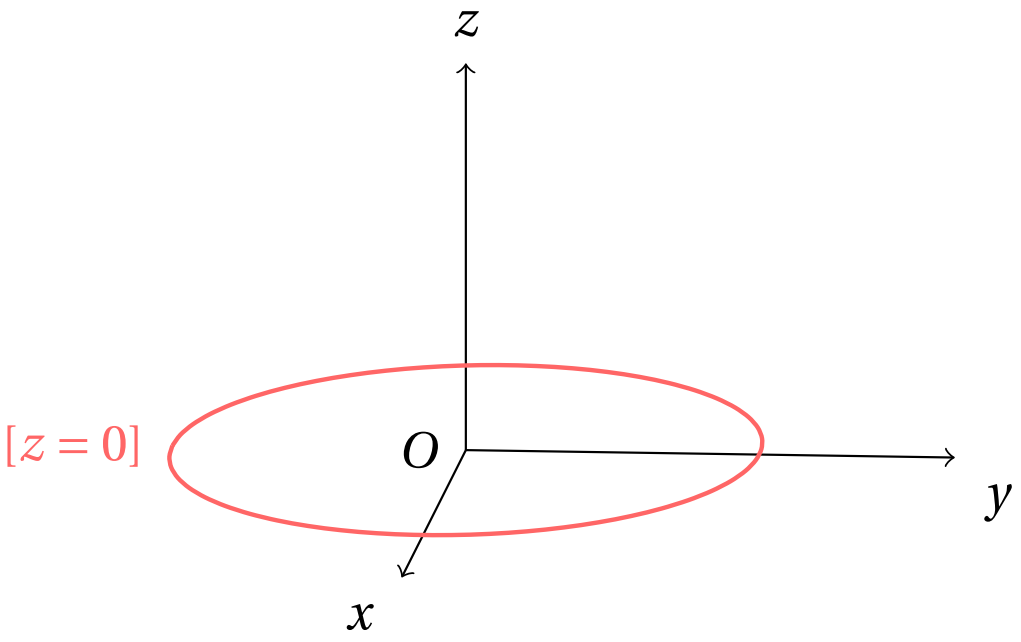

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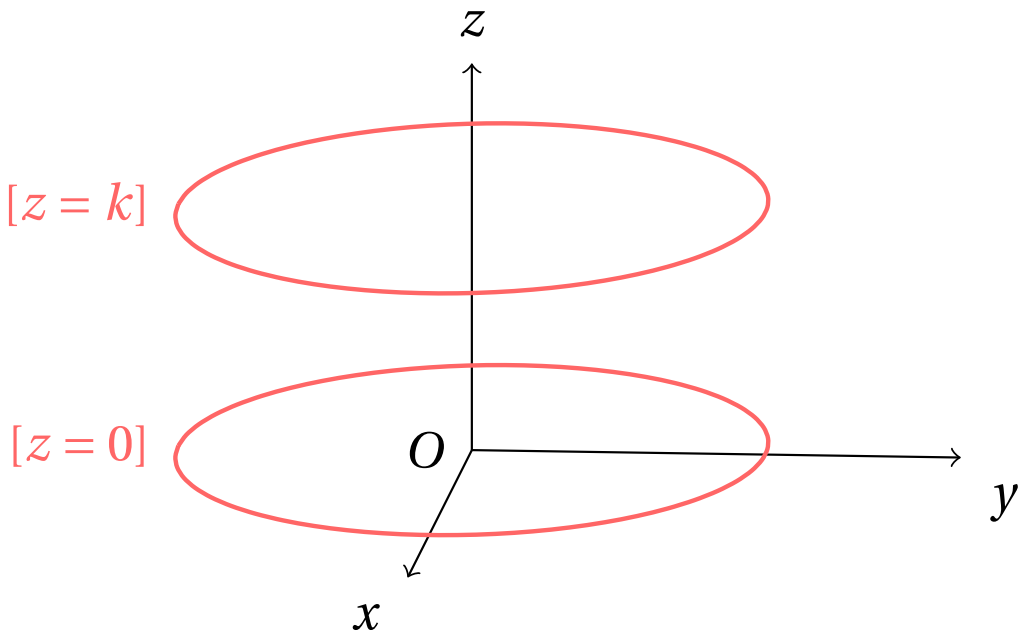

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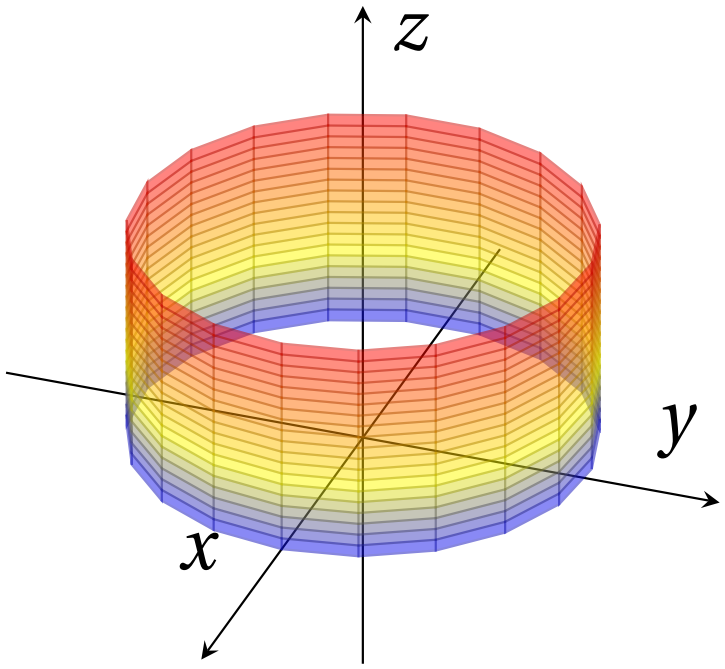

$$\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$$

Mặt trụ tròn: $a = b$

Mặt trụ Parabol

$$y = ax^2$$

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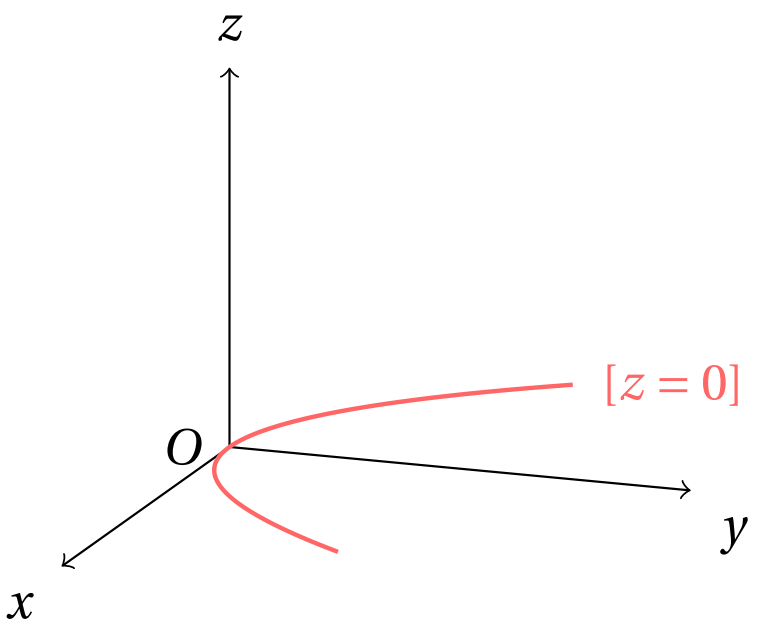

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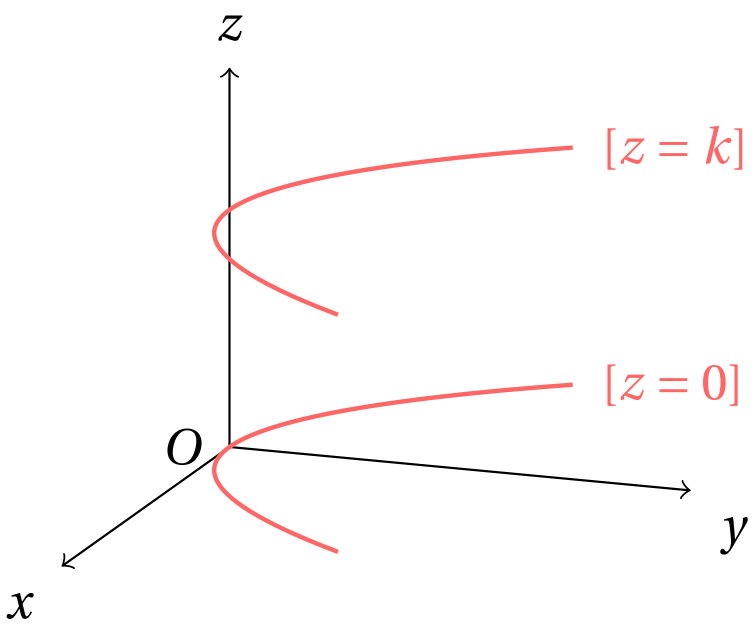

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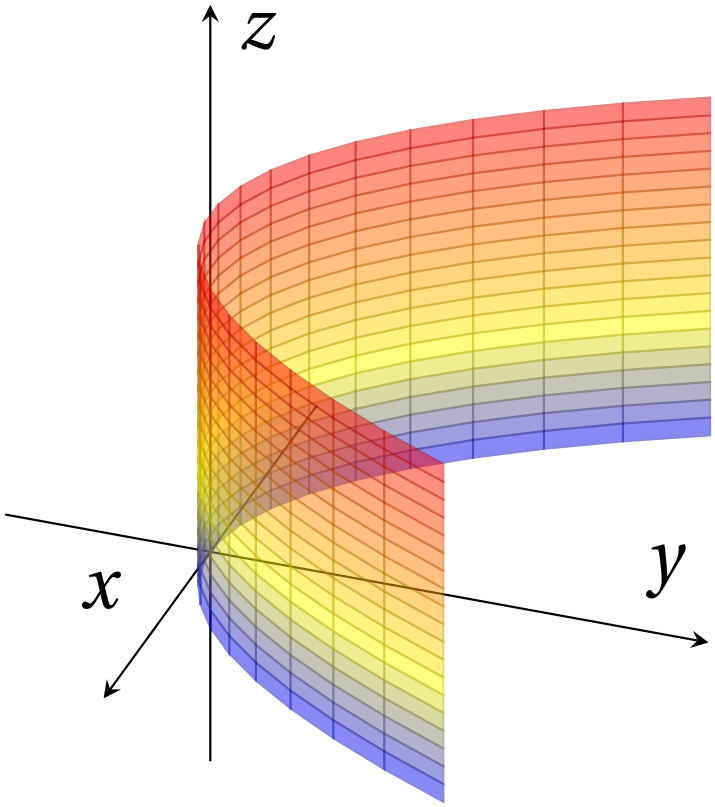

$$y = ax^2$$

Mặt Paraboloid Elliptic

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

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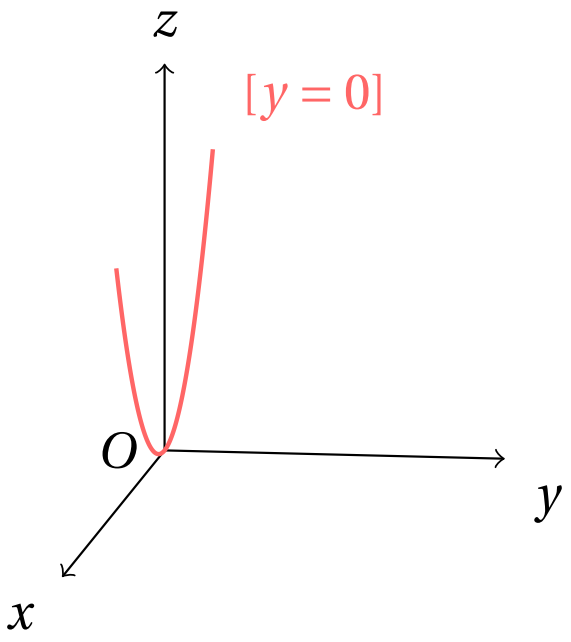

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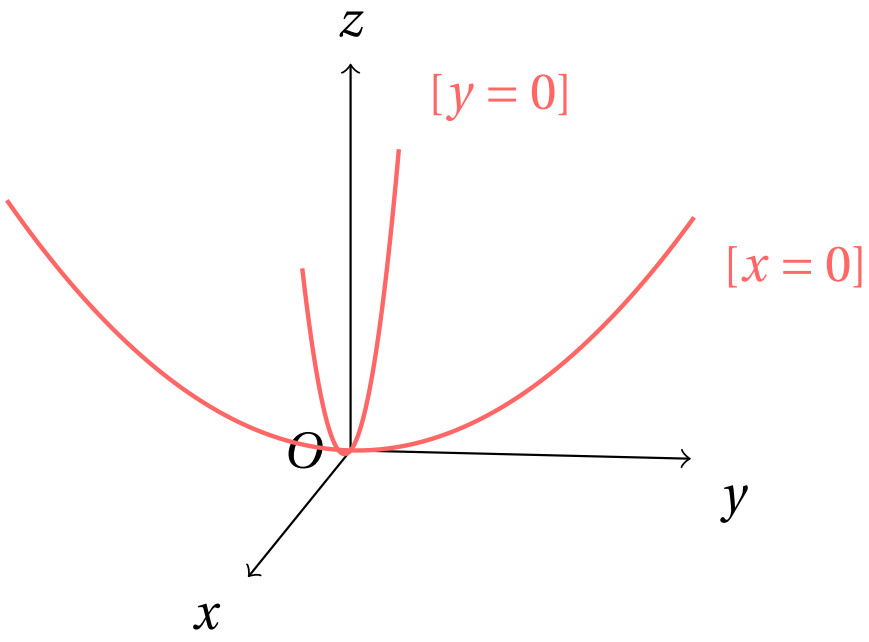

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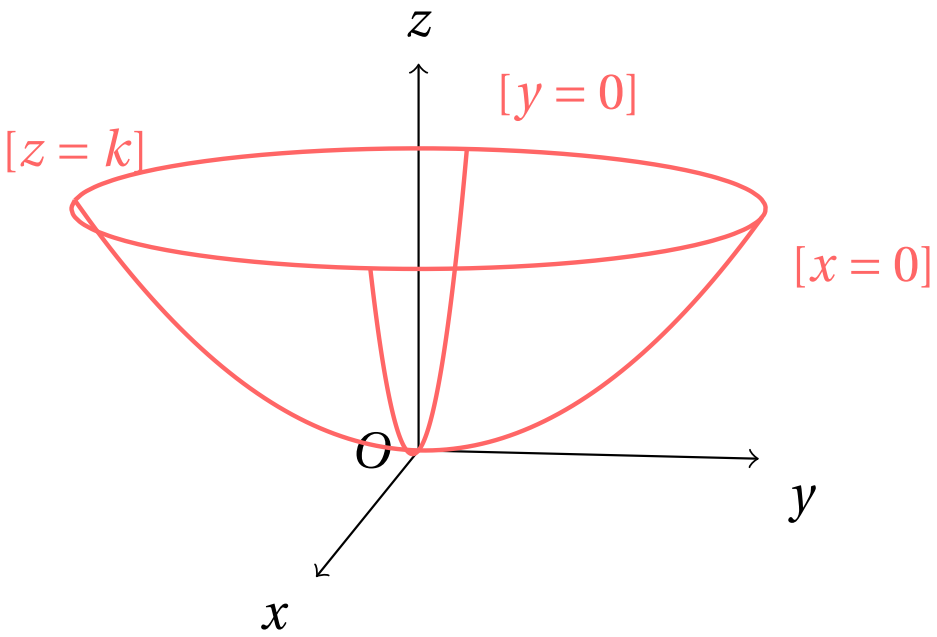

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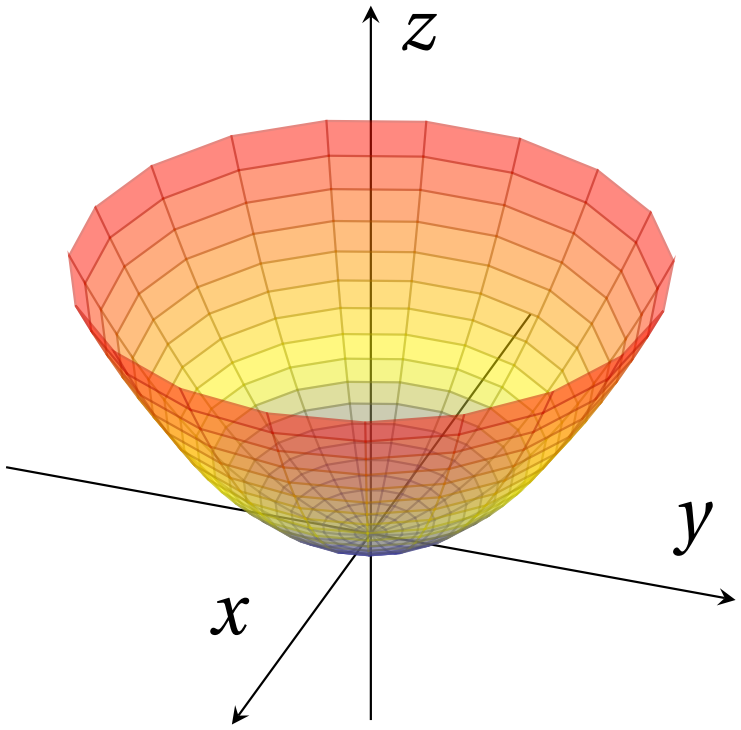

$$z = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

Mặt Paraboloid Hyperbolic

$$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$$

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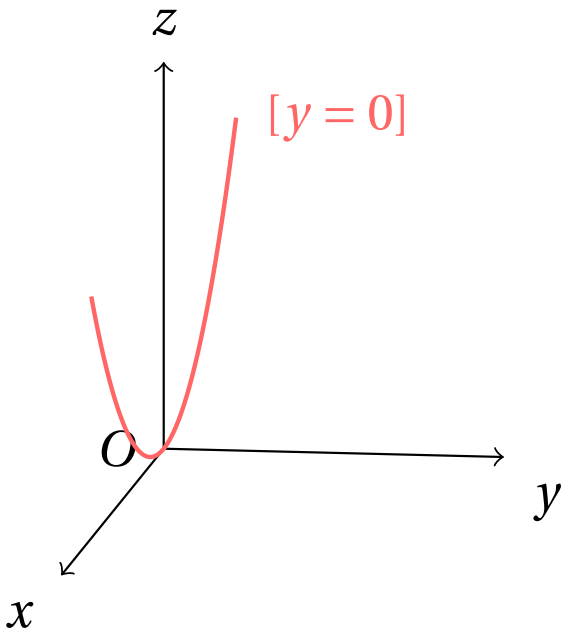

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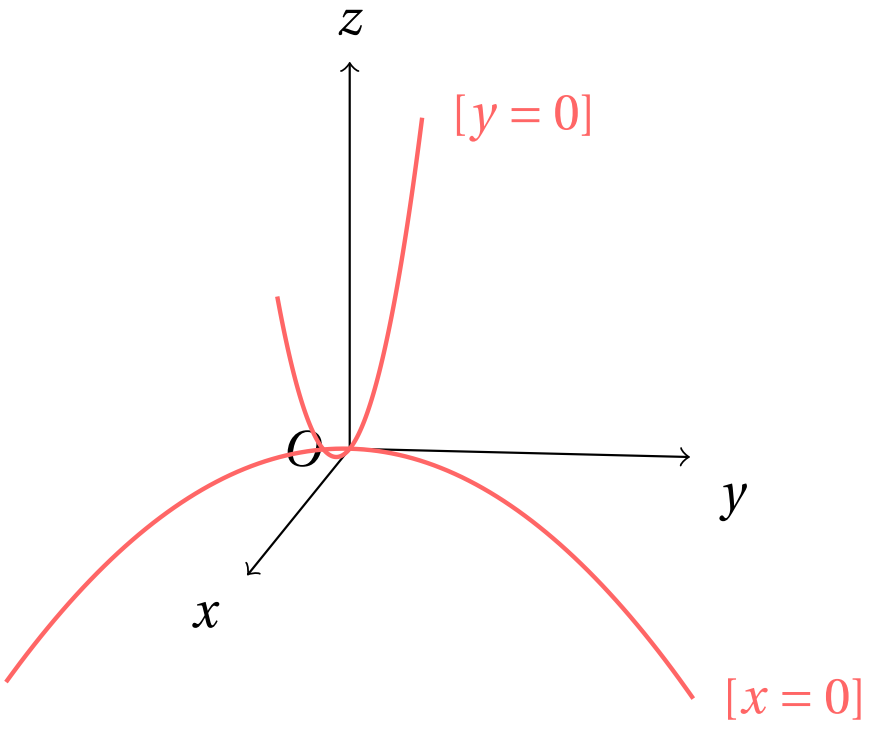

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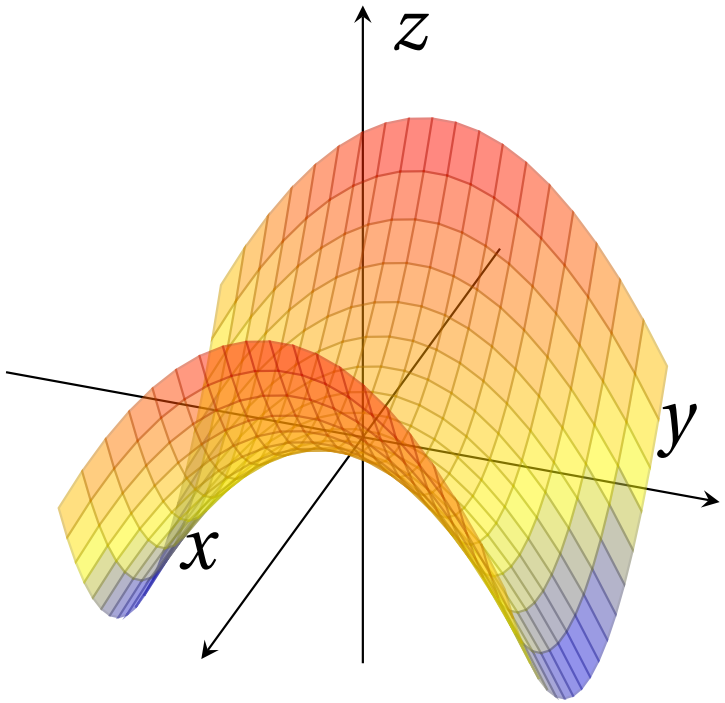

$$z = \frac{x^2}{a^2} - \frac{y^2}{b^2}$$

Mặt nón

$$z = \sqrt{\frac{x^2}{a^2} + \frac{y^2}{b^2}}$$

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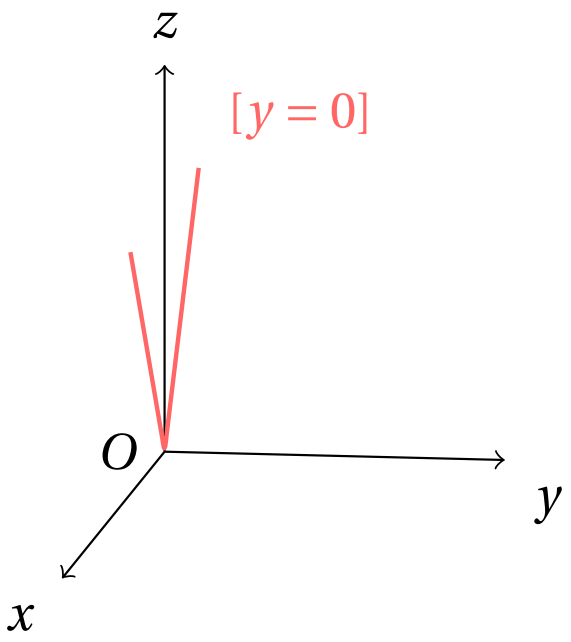

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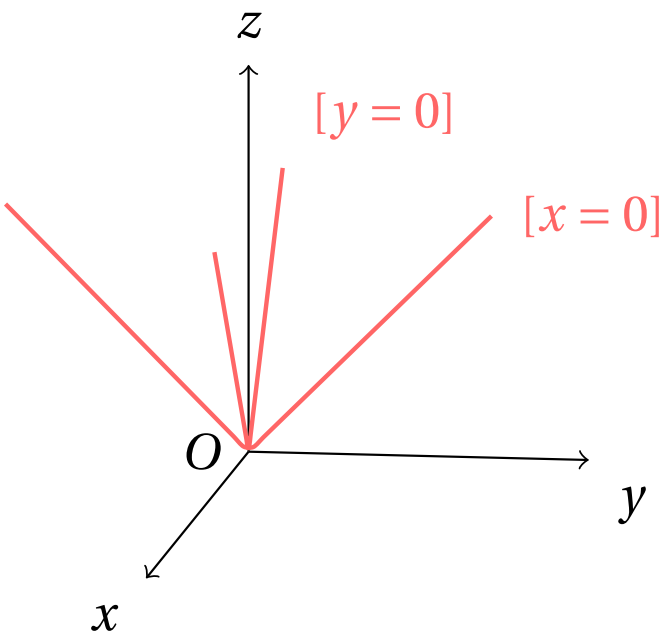

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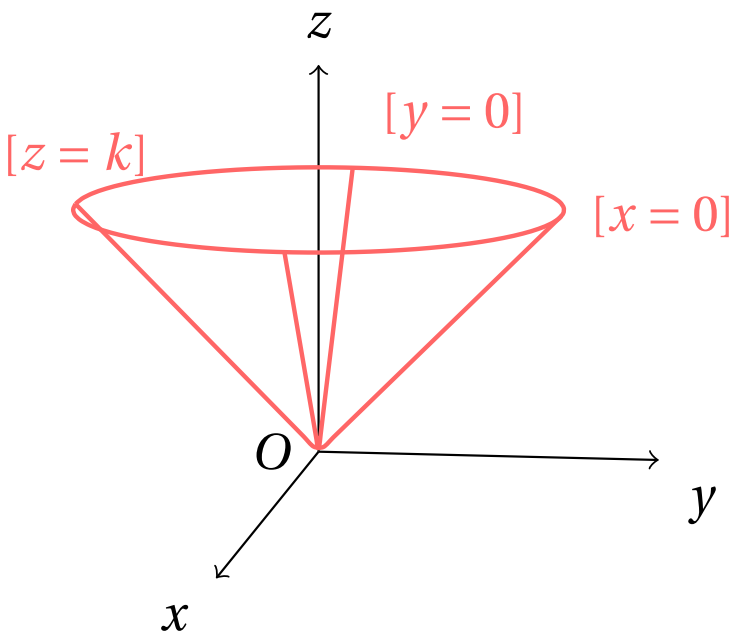

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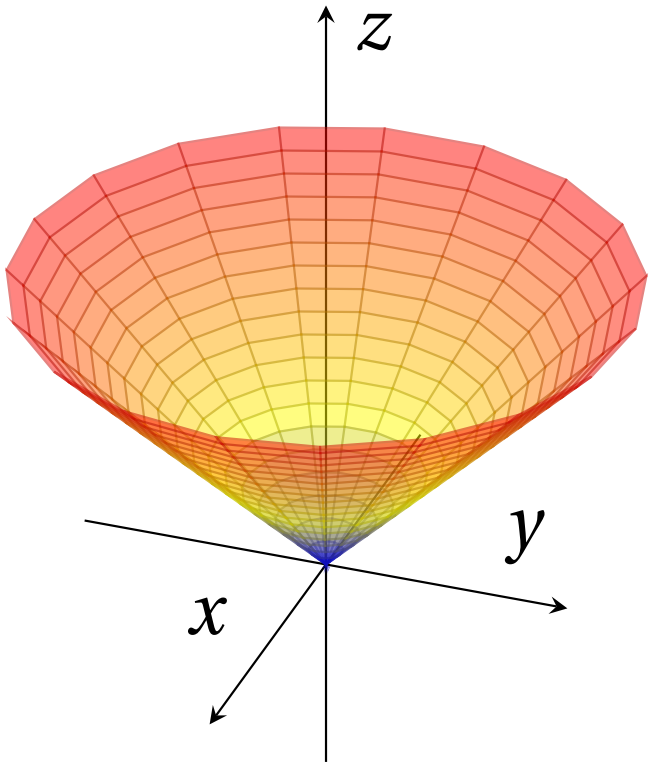
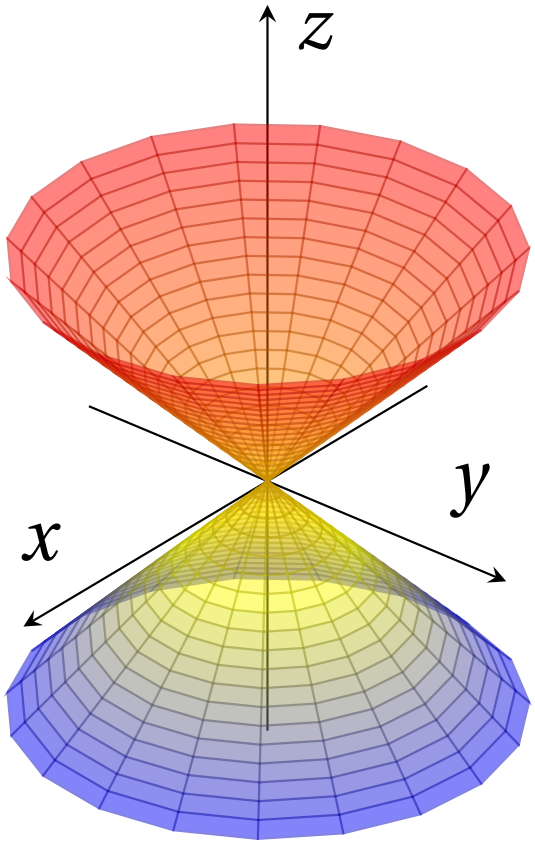


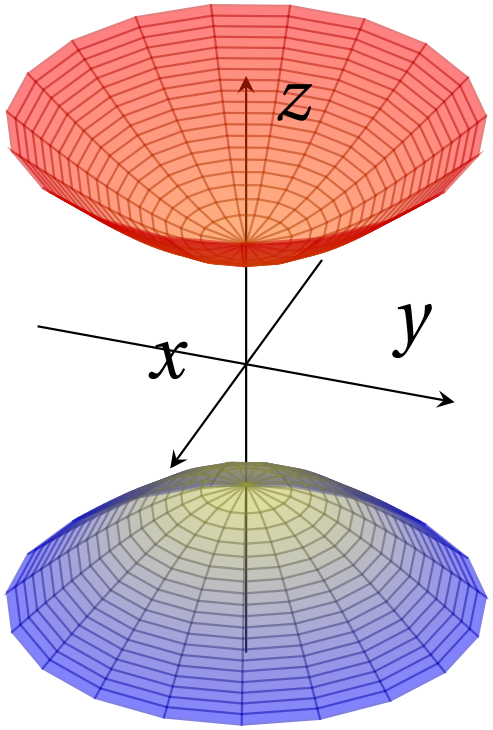
$$z = \sqrt{\frac{x^2}{a^2} + \frac{y^2}{b^2}}$$

Mặt nón hai phía



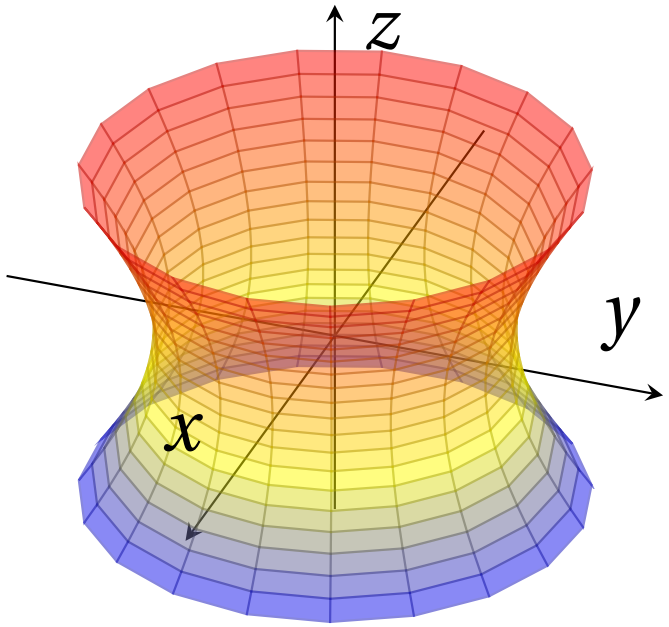
$$z^2 = \frac{x^2}{a^2} + \frac{y^2}{b^2}$$

Mặt Hyperboloid hai tầng



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = -1$$

Mặt Hyperboloid một tầng



$$\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$$

Phát họa để nhận dạng mặt bậc hai: Cho lần lượt x, y, z bằng 0 hoặc bằng $k \neq 0$.